

Some useful LaTeX

FD	Exclude these from S_2 when computing closure	Closure	Decision
1	1	There's no way to get C without this FD	keep
2	2	$A^+ = ACGBEFDH$	discard
3	2, 3	$A^+ = ACBEFHGD$	discard
4	2, 3, 4	$A^+ = ACEFHGDB$	discard
5	2, 3, 4, 5	$A^+ = ACFHGDEB$	discard
6	2, 3, 4, 5, 6	There's no way to get F without this FD	keep
7	2, 3, 4, 5, 7	$BCG^+ = BCG$	keep
8	2, 3, 4, 5, 8	There's no way to get A without this FD	keep
9	2, 3, 4, 5, 9	There's no way to get H without this FD	keep
10	2, 3, 4, 5, 10	Duplicate FD to (14)	discard
11	2, 3, 4, 5, 10, 11	There's no way to get B without this FD	keep
12	2, 3, 4, 5, 10, 12	$H^+ = HEG$	keep
13	2, 3, 4, 5, 10, 13	There's no way to get E without this FD	keep
14	2, 3, 4, 5, 10, 14	There's no way to get G without this FD	keep

A	B	C	D	G	closure	FDs
✓					$A^+ = A$	nothing
	✓				$B^+ = BCDAG$	$B \rightarrow ACDG$
		✓			$C^+ = CAG$	$C \rightarrow AG$: violates BCNF; abort the projection

$\begin{array}{c cc} B & C & D \\ \hline b & 1 & c \\ f & 1 & g \end{array}$	$\begin{array}{c cc} C & A & G \\ \hline 1 & a & d \end{array}$	$\begin{array}{c cc} E & B & F \\ \hline 2 & b & 4 \\ 2 & f & 5 \end{array}$	$\begin{array}{c cc} B & F & H \\ \hline b & 4 & 3 \\ f & 5 & 3 \end{array}$
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Attribute	Appears on		Conclusion
	LHS	RHS	
D	–	–	must be in every key
A	✓	–	must be in every key
E	–	✓	is not in any key
B, C, F	✓	✓	must check